

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

21st November 2024

SA3 - General Insurance

Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)

Total Marks: 100

Q. 1) A large general insurer Guardian General Insurance (“Guardian”) writes both personal and commercial lines business. Around 30% of its portfolio consists of commercial lines and the management aims to expand this segment by launching new and innovative products, as well as scaling up the existing commercial lines business. You are an actuary working for this Company.

- i) You as an actuary have been tasked with creating a financial plan for the commercial line of business that aligns with Company’s financial objectives. Describe how would you go about preparing the plan? [8]

The digital landscape is becoming increasingly complex, marked by the widespread adoption of Internet of Things (IoT) devices, rapid digitalization of businesses, and a thriving e-commerce ecosystem. This expanding digital footprint has also expanded the attack surface for cyber adversaries.

As cyber threats continue to escalate in both frequency and severity, the potential financial impact on businesses can be devastating. Cyber insurance has emerged as an essential tool in the modern enterprise's risk management toolkit. It not only provides a financial safety net in the aftermath of a cyber-attack but also encourages organizations to adopt proactive cybersecurity measures.

- ii) What are the First Party and Third-Party coverages under Cyber insurance? [6]

With the growing need of cyber insurance, the Head of Underwriting has come up with a proposal to introduce retail and commercial cyber products.

- iii) How are the two products- Retail and Commercial Cyber expected to differ? [10]

- iv) Discuss how the different stakeholders/areas which will be impacted or will need attention from the management before the management takes a decision. [10]

The Company has recently launched a drone insurance product. Your friend has currently purchased a drone. He wants to understand the drone insurance product and how should he go about purchasing the apt cover for his drone.

- v) How will you explain this product to him and recommend to him the most appropriate drone insurance coverage for his specific needs and operations? [8]

[42 Marks]

Q. 2) The management team of EverSafe Insurance Group has convened to address some critical issues facing the company. The atmosphere is charged with anticipation as executives prepare to discuss key points that could impact the future of the organization. Your role as an actuary is to provide insights and data to guide the conversation.

- i) An executive from the management pointed out that your company has a notably low claims settlement ratio compared to industry peers. This observation raises concerns and prompts a deeper investigation into the factors contributing to this situation. What various factors according to you might contribute to the difference in claims settlement ratios between industry peers? [7]

- ii) The executives emphasise on having the best practices to maintain a strong solvency ratio as they may go for stock exchange listing. You are called upon to discuss the factors that can

lead to insolvency and the potential consequences if the company fails to address them effectively. [10]

[17 Marks]

Q. 3) You are an actuary at a large general insurance company, "Alpha Insurance," which aims to acquire a smaller competitor, "Beta Insurance." The acquisition aims to expand market share and diversify product offerings.

- i) Discuss the factors to consider when evaluating the reserves of the target company. Furthermore, include considerations for reserve estimation for the combined entity, assuming the merger takes place. [13]

Alpha is also undertaking a separate exercise to produce the financial statements as per IND AS 17. Based on the accounting standard of IND AS 17 for this business, answer the following questions:

- ii) Describe the factors that influence risk adjustment across various lines of business, providing relevant examples. [5]
- iii) Given the following information at initial recognition for a line of business- 'ABC' with 5 years contract of uniform earning:

Premium=INR 5,00,00,000

Expected claims and expenses=INR 4,00,00,000.

What will be the loss component and/or Contractual service margin and how is it expected to be recognised? State any assumptions used. [5]

- iv) How do the IFRS 17 requirements vary for reinsurance contracts from direct contracts? [5]
- v) To streamline this process, the company is considering implementing a vendor tool that automates the calculations and disclosures required under this standard. As the lead actuary, you are tasked with evaluating potential vendor solutions to ensure they meet the company's needs. What are some key considerations which insurer ABC should consider? [5]

[33 Marks]

Q. 4) You are an actuary at "Pinnacle Protectors," tasked with reviewing the pricing of a key general insurance policy. The company is facing increased competition, and accurate pricing is crucial for maintaining profitability. However, you are aware that the data available for this review has some known issues.

Explain why it is important to understand the nature of any flaws, inconsistencies, errors and omissions within a set of data that is to be used for a review of pricing.

[8 Marks]
